

 Marwadi University Marwadi Chandarana Group	Marwadi University Faculty of Engineering & Technology	
Subject: Digital Signal and Image Processing(01CT1513)	To enhance the visual quality of the degraded images using Histogram Equalization and Histogram Matching, and to compare both techniques based on contrast, brightness, and image clarity.	
OPE-8	Date:	Enrollment No:- 92400133138

Aim:- To enhance the visual quality of the degraded images using Histogram Equalization and Histogram Matching, and to compare both techniques based on contrast, brightness, and image clarity.

Code :-

```
import cv2
import numpy as np
import matplotlib.pyplot as plt

def generate_histogram(image):
    #calculate the histogram of the grayscale image
    histogram = cv2.calcHist([image], [0], None, [256], [0, 256])
    return histogram

input_path = r"C:\Users\dhruv\Pictures\Screenshots\Screenshot 2024-05-21 121309.png"
input_image = cv2.imread(input_path)
gray_image = cv2.cvtColor(input_image, cv2.COLOR_BGR2GRAY)

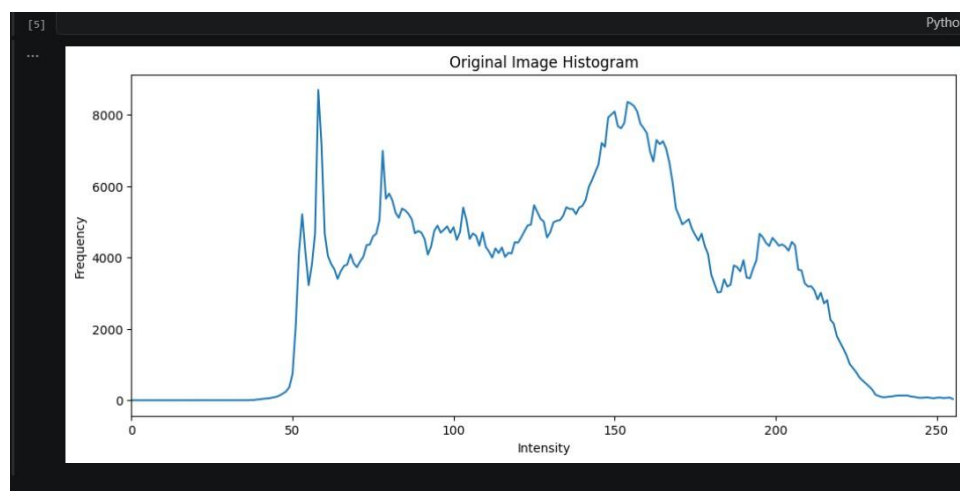
input_hist = generate_histogram(gray_image)

intensity = np.arange(256)

plt.figure(figsize=(12, 5))

plt.plot(intensity, input_hist)
plt.title("Original Image Histogram")
plt.xlabel("Intensity")
plt.ylabel("Frequency")
plt.xlim([0, 256])
plt.show()
```

OUTPUT:-



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Code :-

```
# display the grayscale image
plt.subplot(1, 2, 1)
plt.imshow(gray_image, cmap='gray')
plt.title("input Image")
plt.axis('off')
```

OUTPUT:-

```
(np.float64(-0.5), np.float64(913.5), np.float64(913.5), np.float64(-0.5))
```

input Image

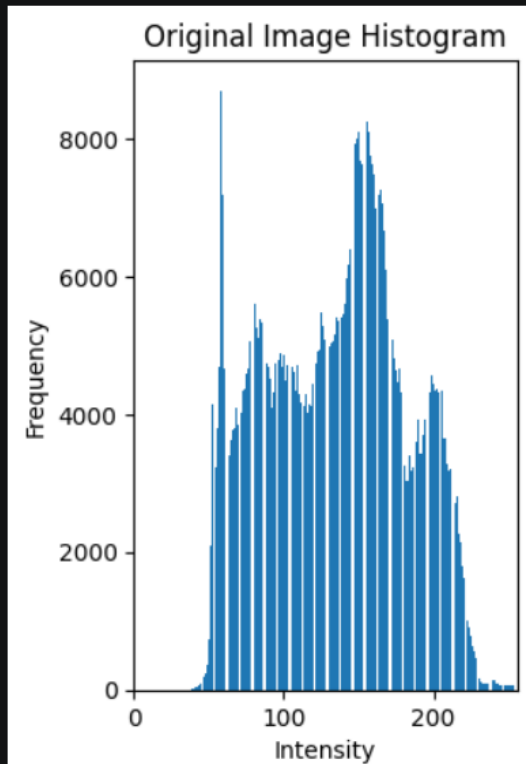


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Code :-

```
# Display the histogram
plt.subplot(1, 2, 1)
plt.bar(intensity, input_hist)
plt.title("Original Image Histogram")
plt.xlabel("Intensity")
plt.ylabel("Frequency")
plt.xlim([0, 256])
plt.show()
```

Output :-



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Code :-

```
# calculate the total number of pixels
total_pixels = gray_image.size

#normalize the histogram
normalized_hist = input_hist / total_pixels

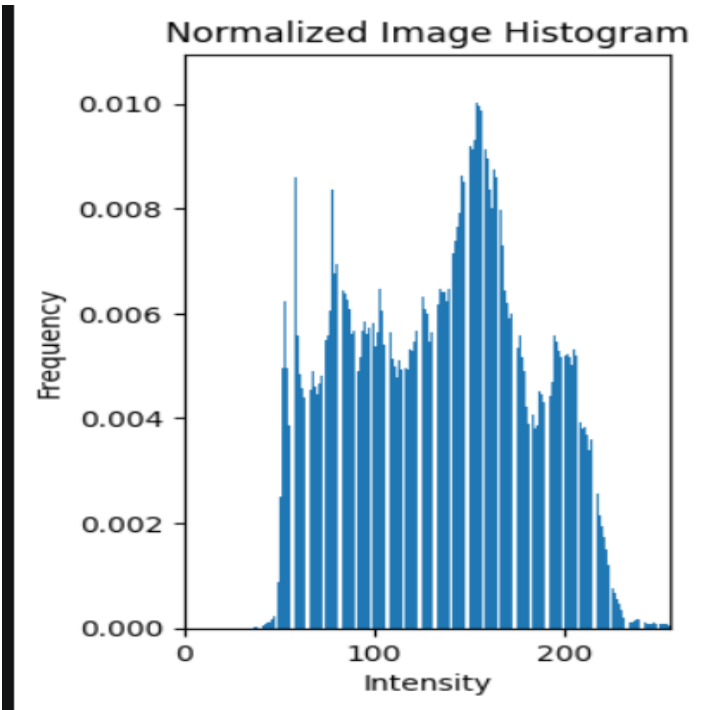
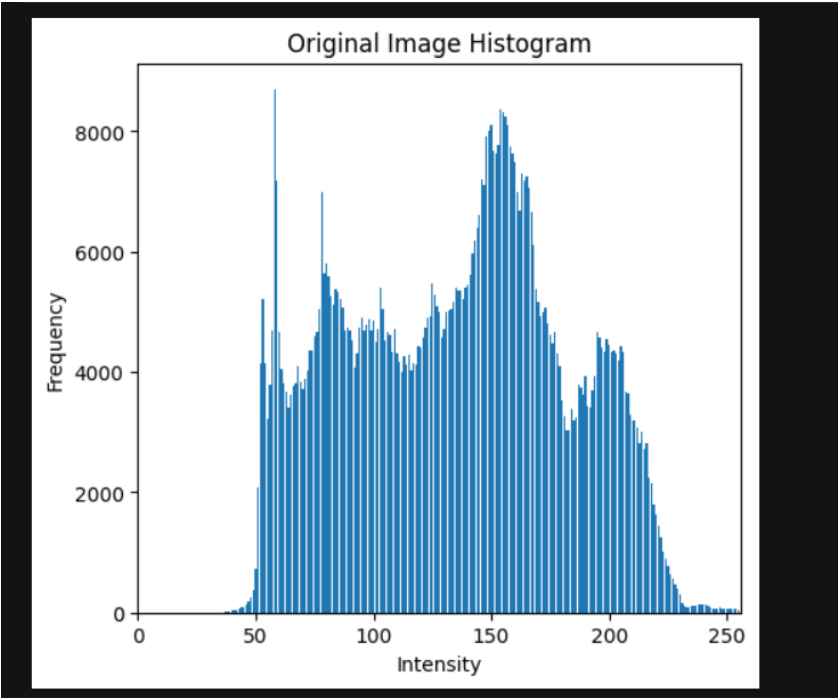

# Display original the normalized histogram
plt.figure(figsize=(12, 5))

#original histogram
plt.subplot(1, 2, 1)
plt.bar(intensity, input_hist)
plt.title("Original Image Histogram")
plt.xlabel("Intensity")
plt.ylabel("Frequency")
plt.xlim([0, 256])
plt.show()

#normalized histogram
plt.subplot(1, 2, 2)
plt.bar(intensity, normalized_hist)
plt.title("Normalized Image Histogram")
plt.xlabel("Intensity")
plt.ylabel("Frequency")
plt.xlim([0, 256])
plt.show()
```

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Conclusion :-

Histogram equalization improves the overall contrast of degraded images by spreading pixel intensities across a wider range. Histogram matching adjusts the image histogram to resemble a reference image, producing more controlled and natural-looking enhancement. The better technique depends on the image: equalization works well when strong global contrast enhancement is needed, while histogram matching is better when a specific visual appearance or reference contrast is desired.

GITHUB LINK:-

<https://github.com/ayush-subudhi/Dsip-Experiment>